



28-MAVZU · III BOB. PYTHON VA MA'LUMOTLAR BAZASI

Ma'lumotlarni qo'shish, ko'rish va saqlash. CRUD operatsiyalari

CRUD konsepsiyasi; INSERT bilan qo'shish, SELECT bilan o'qish (ustunlarni tanlash, WHERE, ORDER BY, LIMIT), UPDATE bilan yangilash, DELETE bilan o'chirish; WHERE'siz UPDATE/DELETE xavfi; o'quvchilar bazasi loyihasi.

Bugun nimani o'rganamiz?

- **CRUD** — har qanday dasturning 4 asosiy amali
- **INSERT** — qo'shish, **SELECT** — o'qish
- **UPDATE** — o'zgartirish, **DELETE** — o'chirish
- Eng xavfli xato: **WHERE'siz** UPDATE/DELETE
- O'quvchilar bazasi (18-loyiha)

CRUD nima?

Harf	Amal	SQL	Instagram'da	HTTP
Create	Yaratish	INSERT	Post joylash	POST
Read	O'qish	SELECT	Lentani ko'rish	GET
Update	Yangilash	UPDATE	Izohni tahrirlash	PUT/PATCH
Delete	O'chirish	DELETE	Postni o'chirish	DELETE

- 27-mavzudagi **maktab** bazasi bilan ishlaymiz

Create — INSERT

SQL

```
INSERT INTO sinflar (nomi, sinf_rahbari, xona)
VALUES ('8-A', 'Karimova Nodira', 204);
```

```
INSERT INTO sinflar (nomi, sinf_rahbari, xona) VALUES
('8-B', 'Rasulov Akmal', 206),
('8-V', 'Yusupova Malika', 210);
```

```
INSERT INTO oquvchilar (ism, familiya, tugilgan_sana, email, sinf_id) VALUES
('Ali', 'Valiyev', '2012-03-15', 'ali@mail.uz', 1),
('Zarina', 'Karimova', '2012-07-02', 'zarina@mail.uz', 1),
('Sardor', 'Qodirov', '2012-05-30', NULL, 3);
```

- Ustunlar va qiymatlar **bir xil tartib va son**da
- **id** yozilmaydi — **SERIAL** o'zi beradi; **NULL** — "qiymat yo'q"

Baza cheklovlarni tekshiradi

SQL

```
INSERT INTO oquvchilar (familiya) VALUES ('Aliyev');  
-- ✗ null value in column "ism" violates not-null constraint
```

```
INSERT INTO oquvchilar (ism, familiya, email) VALUES ('Vali', 'Aliyev', 'ali@mail.uz');  
-- ✗ duplicate key value violates unique constraint
```

```
INSERT INTO oquvchilar (ism, familiya, sinf_id) VALUES ('Vali', 'Aliyev', 99);  
-- ✗ violates foreign key constraint
```

```
INSERT INTO baholar (oquvchi_id, fan, baho) VALUES (1, 'Matematika', 7);  
-- ✗ violates check constraint
```

Bu xatolar — yaxshi! Baza noto'g'ri ma'lumotga yo'l qo'ymaydi

Read — SELECT

SQL

```
SELECT * FROM oquvchilar;           -- hammasi
SELECT ism, familiya FROM oquvchilar; -- kerakli ustunlar

SELECT ism || ' ' || familiya AS toliq_ism,      -- || birlashtirish, AS nom
       EXTRACT(YEAR FROM AGE(tugilgan_sana)) AS yoshi
FROM oquvchilar;
```

Natija

toliq_ism		yoshi
-----+-----		
Ali Valiyev		14
Zarina Karimova		14

WHERE — shart bo'yicha tanlash

SQL

```
SELECT * FROM oquvchilar WHERE sinf_id = 1;
SELECT * FROM oquvchilar WHERE ism = 'Bobur';
SELECT * FROM oquvchilar WHERE tugilgan_sana < '2012-01-01';
SELECT * FROM oquvchilar WHERE sinf_id = 1 AND alochi = TRUE;
SELECT * FROM oquvchilar WHERE sinf_id = 1 OR sinf_id = 2;
SELECT * FROM oquvchilar WHERE email IS NULL;      -- = NULL emas!
```

- Operatorlar: = <> < > ≤ ≥, AND OR NOT
- LIKE, IN, BETWEEN — 30-mavzuda

ORDER BY va LIMIT

SQL

```
SELECT ism, familiya FROM oquvchilar ORDER BY familiya;           -- A → Z
SELECT ism, tugilgan_sana FROM oquvchilar ORDER BY tugilgan_sana DESC;
SELECT ism FROM oquvchilar ORDER BY sinf_id, familiya;           -- ikki daraja
SELECT * FROM oquvchilar ORDER BY id LIMIT 3;                     -- birinchi 3 ta
SELECT * FROM oquvchilar ORDER BY id LIMIT 3 OFFSET 3;            -- 2-sahifa
```

SELECT → FROM → WHERE → ORDER BY → LIMIT

- Tartib **qat'iy**: **WHERE** ni **ORDER BY** dan keyin yozib bo'lmaydi

Update — UPDATE

SQL

```
UPDATE oquvchilar  
SET email = 'sardor@mail.uz'  
WHERE id = 5;
```

```
UPDATE oquvchilar  
SET alochi = TRUE, telefon = '+998901234567'  
WHERE id = 2;
```

```
UPDATE sinflar SET xona = xona + 100 WHERE nomi = '8-A';
```

```
UPDATE oquvchilar SET sinf_id = 2 WHERE id = 6 RETURNING *;
```

Delete va ENG XAVFLI XATO

SQL

```
DELETE FROM oquvchilar WHERE id = 7;  
DELETE FROM baholar WHERE sana < '2026-01-01';  
  
UPDATE oquvchilar SET sinf_id = 1;      -- 😱 BARCHA o'quvchilar 8-A ga!  
DELETE FROM oquvchilar;                 -- 😱 BARCHA o'quvchilar o'chdi!
```

⚠ Baza "Ishonchingiz komilmi?" deb so'ramaydi

Avval xuddi shu WHERE bilan SELECT — keyin UPDATE/DELETE

SQL

```
SELECT * FROM baholar WHERE sana < '2026-01-01';      -- 1. tekshiring  
DELETE FROM baholar WHERE sana < '2026-01-01';        -- 2. keyin o'chiring
```

Tranzaksiya — "hammasi yoki hech biri"

SQL

```
BEGIN;  
UPDATE oquvchilar SET sinf_id = 3 WHERE id = 1;  
INSERT INTO baholar (oquvchi_id, fan, baho) VALUES (1, 'Matematika', 5);  
COMMIT;                -- hammasini saqlash  
-- Xato bo'lsa: ROLLBACK; - hammasi bekor qilinadi
```



Pul o'tkazmasi: sizdan yechiladi **VA** do'stingizga tushadi. Orada elektr o'chsa — pul "yo'qolmasligi" uchun ikkalasi birga bajariladi yoki ikkalasi bekor qilinadi.

BO'LIM

Amaliy mashg'ulot

18-loyiha: o'quvchilar bazasi

Amaliyot: topshiriqlar

- 1 Sinflar, o'quvchilar (+5 sinfdosh) va **20+ baho** qo'shing
- 2 So'rovlar:
 - To'liq ism, familiya bo'yicha tartibda
 - 2-sinf o'quvchilari
 - Eng yoshi katta 3 nafar
 - Emaili kiritilmaganlar
 - Informatikadan 5 olganlar
- 3 Sardorga email qo'shing; Zarinani a'lochi qiling
- 4 Bitta o'quvchini o'chiring (avval SELECT!) — baholari ham o'chdimi? Nega?

Amaliyot: javoblar

SQL

```
SELECT ism || ' ' || familiya AS toliq_ism FROM oquvchilar ORDER BY familiya;  
SELECT * FROM oquvchilar WHERE sinf_id = 2;  
SELECT ism, familiya, tugilgan_sana FROM oquvchilar ORDER BY tugilgan_sana LIMIT 3;  
SELECT ism, familiya FROM oquvchilar WHERE email IS NULL;  
SELECT * FROM baholar WHERE fan = 'Informatika' AND baho = 5;
```

- 4-savol javobi: **ON DELETE CASCADE** — baholar ham avtomatik o'chadi

Ko'p uchraydigan xatolar

- ❌ `WHERE email = NULL` — hech narsa qaytarmaydi; `IS NULL`
- ❌ `WHERE ism = Ali` — qo'shtirnoqsiz: `column "ali" does not exist`
- ❌ Ustunlar va qiymatlar soni mos emas
- ❌ **WHERE'siz** UPDATE/DELETE — avval SELECT!

Savol

CRUD'dagi "R" qaysi SQL buyrug'iga mos keladi?

A INSERT

B SELECT

C UPDATE

D REMOVE

Javobni tanlang — to'g'ri variant yashil bo'ladi

Savol

DELETE FROM oquvchilar; nima qiladi?

A Faqat birinchi qatorni o'chiradi

B Jadvalni o'chiradi

C Xato beradi

D Jadvaldagi BARCHA qatorlarni o'chiradi

Javobni tanlang — to'g'ri variant yashil bo'ladi

Savol

Emaili kiritilmagan o'quvchilarni topuvchi to'g'ri shart?

A WHERE email IS NULL

B WHERE email = NULL

C WHERE email = ''

D WHERE NOT email

Javobni tanlang — to'g'ri variant yashil bo'ladi

Savol

Natijani kamayish tartibida saralash uchun?

A ORDER BY ... ASC

B SORT BY ... DOWN

C ORDER BY ... DESC

D ORDER DESC BY ...

Javobni tanlang — to'g'ri variant yashil bo'ladi



Uyga vazifa

- 1 `dokon` bazasini to'ldiring: 3 toifa, 15 mahsulot, 5 mijoz
- 2 Eng qimmat 5 ta; tugaganlar (`soni = 0`); 50 000 dan arzonlar alifbo tartibida
- 3 "Kanselyariya" narxini 10% oshiring: `SET narx = narx * 1.1` — **avval SELECT!**

Yodda tuting

- CRUD: `INSERT`, `SELECT`, `UPDATE`, `DELETE`
- `INSERT INTO t (a, b) VALUES (1, 'x');`
- `SELECT ... FROM t WHERE ... ORDER BY u DESC LIMIT n;`
- `UPDATE ... WHERE`, `DELETE ... WHERE` — WHERE'ni unutmang, avval SELECT!
- `IS NULL`; tranzaksiya: `BEGIN ... COMMIT / ROLLBACK`



Savollar bormi?

Keyingi mavzu: **29. Jadvallarni bog'lash va JOIN. Index, constraint, data integrity. Backup va restore**

Batafsil matn, misollar va testlar: **28-sql-crud.html** darsida